Feng-Yuan “Frey” Liu

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Research Interests

Galaxy formation & evolution, IR/(sub)mm/radio observation, star-forming galaxies, dust & gas at high redshift, first galaxies, active galactic nuclei, gravitational lensing, large-scale structure, cosmic reionization

Education

Ph.D. in Astronomy — *Sep. 2023 – present (expected 4 years)*
Institute for Astronomy (IfA), University of Edinburgh (UoE) — Edinburgh, UK

- Mary Brück scholarship. Supervisors: Prof. **James Dunlop**, Prof. **Ross McLure**, Dr. **Derek McLeod**
- Thesis: *Galaxy Evolution Across Cosmic Time Through Multi-Wavelength Observations*

M.Sc. in Astrophysics — *Sep. 2020 – Jun. 2023*
National Astronomical Observatories, Chinese Academy of Sciences (NAOC) — Beijing, China

- Class Valedictorian. Supervisor: Prof. **Yu Sophia Dai**
- Thesis: *Cold gas and Dust Properties of Infrared Luminous Quasars*

B.Sc. in Astronomy — *Sep. 2016 – Jun. 2020*
School of Astronomy and Space Science, Nanjing University (NJU) — Nanjing, China

- Astronomy Elite Class (summa cum laude equivalent). Supervisor: Prof. **Yi Xie**
- Thesis: *Gravitational Lensing of a Regular Non-Minimal Einstein-Yang-Mills Black Hole*

Telescope Programs **(total approved time as PI: ∼420 hr)**

JWST:

1. **Co-I** (GO 11171, PI: D. McLeod; co-PI: N. Garuda) — Cycle 5, 2026, **12.3h**
   Is the quest for Cosmic Dawn beyond all H0pe?

VLA:

1. **First author** (26B-312, **Large Program**, **Priority A**) — 2026B, **157.43h**
   THISTLE: A Deep VLA Legacy Survey Around the Abell 370 Cluster

NOEMA (total: **273h**):

6. **PI** (S26CO, co-PI: P. Cox) — 2026S, **64.0h**
   NOSEGAY: The Northern-Sky Submillimeter Galaxy Census at the Final CANDELS Frontier
5. **PI** (S26DE, co-PI: P. Cox) — 2026S, **43.5h**
   NACRE-UDS: A Cosmic-Variance-Mitigated View of Dusty Star-Forming Galaxy Emergence at the EoR
4. **PI** (W25EC, co-PI: P. Cox) — 2025W, **86.4h**
   NACRE: Tracing the early emergence of dusty star-forming galaxies with continuum and [CII] emission
3. **PI** (S25CK, co-PI: P. Pérez-González, **Grade A**) — 2025S, **65.0h**
   Tracing the early emergence of dusty star-forming galaxies at the Epoch of Reionization
2. **Co-I** (W24DQ, PIs: Q. Tan, D. Liu) — 2024W, **4.0h**
   Hyper-luminous infrared galaxies exhibiting narrow CO lines: strong lenses or genius extreme starburts?
1. **Co-I** (S22CH, PI: Y. S. Dai) — 2022S, **7.3h**
   Confirmation of rare starburst HyLIRGs in high-z QSOs

GMRT:

1. **Co-I** (No. 42-012, PI: C. He) — Cycle 42, 2022, **20.0h**
   Spatially Resolved HI Kinematics of Spiral+Elliptical and Spiral+Spiral Major-merger Galaxy Pairs

Selected Grants, Scholarships & Honors (total: ~136,000 USD)

Summer project bursary grant (£1,500), Royal Astronomical Society (RAS)	*May 2026*
Best Poster Prize, PhD Highlands Conference, School of Physics and Astronomy (SoPA), UoE	*Feb. 2026*
Travelling Grant for Telescope Service (£2,065), IfA, UoE	*Jan. 2026*
Best Talk Prize (£50), DEX-XXII	*Jan. 2026*
Travelling Grant (£3,298), IfA, UoE	*Sep. 2025*
CSE Vacation Funding (**£1,709**, co-applied as supervisor), UoE	*Jun. 2025*
Fellow of the Royal Astronomical Society	*May 2025*
Mary Brück Scholarship (£20,780 per annum for 4 years), UoE	*2023–2027*
Research Training Support Grant (£4,800), UoE	*Sep. 2023*
Valedictorian of Class 2023, NAOC	*Jun. 2023*
National Graduate Student Award (20,000 CNY), Ministry of Education of PRC	*Oct. 2022*
First Class Scholarship for Master's Students (43,200 CNY per annum for 3 years), NAOC	*2020–2023*
First Award of Lin-bridge Scholarship (3,000 CNY), Kavli Institute for Astrophysics, Peking University	*Sep. 2020*
First Award for Best Undergraduate Thesis, NJU	*Jun. 2020*
Selected member for National Top Notch honors track, Ministry of Education of PRC	*Sep. 2017 – Jun. 2020*

Research Skills

Observation: ESO visiting astronomer, Danish 1.54m telescope at La Silla, Chile, 20-night service (Jun. & Sep. 2025) for Milky Way + solar system targets & main author of a new focusing strategy

Proposal writing: VLT (X-SHOOTER, MOONS), Keck (MOSFIRE), JWST (NIRCam + NIRSpec), ALMA, NOEMA, VLA (pointed+mosaic), GMRT

Data reduction & analysis: SDSS, DK154, VLT/MOONS, Keck/MOSFIRE, JWST/NIRCam, VISTA, HST, ALMA, NOEMA, IRAM-30m, GMRT

Proficient software: `CASA, GILDAS, DS9, TOPCAT, CARTA, SExtractor, CIGALE, Bagpipes, Maple`

Code releases (Python-based projects): `APRICOT` (expected 2026)

Languages: English (fluent), Chinese Mandarin (native)

Teaching, Supervising & Professional Service

Supervisor, Summer Research Program — Edinburgh, UK, *Jun. – Jul. 2026 (expected)*
- Student: Daria Nemova
- Project: *Molecular Line Emission from Typical High-Redshift Galaxies*
- Funded (£1,500) by RAS Undergraduate Bursary

Primary supervisor, Summer Research Program — Edinburgh, UK, *Jun. – Jul. 2026 (expected)*
- Student: Lewis McNeil (secondary supervisor: Prof. J. Dunlop)
- Project: *Probing the star-formation fuels of massive galaxies in lensing fields with VLT, JWST, and ALMA*
- Funded (£2,152) by Carnegie Undergraduate Vacation Scholarship

Primary supervisor, Summer Research Program — Edinburgh, UK, *Jul. – Aug. 2025*
- Student: Chang Sui (secondary supervisor: Prof. J. Dunlop)
- Project: *A stacking analysis to probe faint (sub)mm galaxies with archival ALMA and JWST data*
- Funded (£1,709) by College of Science and Engineering (CSE), UoE

Chair of Organising Committee, ESEA-I — Edinburgh, UK, *Jun. 2025*
- First Edinburgh School of Extragalactic Astronomy: five days, ~ 30 lectures, ~120 UK & international attendees

Teaching assistant, *Observational Astronomy Theory & Experiment* — Turkana Basin Institute, Kenya, *Sep. 2024*
- Tutoring postgraduate-level training for two weeks (voluntary service for the DARA program)
- Duties incl. operating a 60cm telescope, writing pipelines & hosting data reduction workshops

Distributed peer reviewer for Call for Proposals: ALMA (2024, 2026), ESO VLT (2024)

Selected Conference Presentations

Invited talk:

2. **Royal Observatory Colloquium**, Edinburgh, UK — *May 2026*
   The Dark Side of the Universe: Dusty Star-Forming Galaxies in the Era of JWST and Radio Interferometers
1. **ESEA-I** (summer school), Edinburgh, UK — *Jun. 2025*
   Lecturer for 4 sessions: *Far-infrared, (sub)millimeter and radio emission of distant galaxies*; *Paper reading and management*; *Observation and telescope proposal writing*; *A brief introduction to radiation transport*.

Contributed talks:

9. **Tokyo JWST Conference 2026**, Tokyo, Japan — *Mar. 2026*
   Uncovering the Dust-Obscured Cosmic Star Formation History back to $z\sim7$ with JWST PRIMER and ALMA in COSMOS
8. **DEX-XXII (best-talk prize)**, Edinburgh, UK — *Jan. 2026*
   Uncovering the Dust-Obscured Cosmic Star Formation History back to $z\sim7$ with JWST PRIMER and ALMA in COSMOS
7. **NAM2025**, Durham, UK — *Jul. 2025*
   Dust-obscured Cosmic Star-Formation History Back to $z=7$ Revealed by ALMA and JWST PRIMER
6. **CFC2025**, Austin, US — *May 2025*
   Revealing the Obscured Cosmic Star Formation History at $z=1-7$ with JWST PRIMER and ALMA
5. **DEX-XXI**, Newcastle, UK — *Jan. 2025*
   Unveiling Dusty Star Formation Across Cosmic Time with JWST PRIMER and ALMA
4. **The 2nd Edinburgh-Copenhagen Joint Workshop**, Copenhagen, Denmark — *Oct. 2024*
   Dust-Obscured Star Formation in the Young Universe Revealed by Combining JWST PRIMER and ALMA
3. **DEEP2024**, Sintra, Portugal — *Oct. 2024*
   Dust-Obscured Star Formation in the Young Universe Revealed by Combining JWST PRIMER and ALMA
2. **NAM2024 (two talks)**, Hull, UK — *Jul. 2024*
   Dust-Obscured Star Formation in the Young Universe Revealed by Combining JWST PRIMER and ALMA
   Quasars Co-existing with Dusty Starburst HyLIRGs at the Cosmic Noon
1. **DEX-XX**, Durham, UK — *Jan. 2024*
   Cold Gas and Dust Properties of Infrared-Luminous QSOs

Contributed posters:

2. **The 40th IAP Symposium**, Paris, France — *Dec. 2024*
   Revealing the Obscured Star Formation History of the Universe with JWST PRIMER and ALMA
1. **The XXXII IAU GA**, Cape Town, South Africa — *Aug. 2024*
   Dust and cold gas properties of starburst infrared luminous quasars at $0.5<z<3$

Outreach Highlights

Science writer & translator: 50+ astronomy-related published articles & 5 published books by April 2026

Science influencer (writing + photography + filming): 10M+ reads & 33k+ online followers by April 2026

Public talks: 6 large popular-astronomy talks in English & Chinese as invited speaker by April 2026

Podcasts: 6 episodes as invited speaker, total listens 10k+ by April 2026

Publications (total: 15 h-index: 5 citations: 340+)

See ADS for all published papers

First-authored papers: 4

5. **Liu, F.-Y.**; Sui, C.; Dunlop, J. S.; et al., in prep. — 2026 (expected)
   APRICOT: The first systematic ALMA detections of very faint dust emission in individual galaxies across the JWST/PRIMER COSMOS field

4. **Liu, F.-Y.**; Dunlop, J. S.; McLure, R. J.; et al., MNRAS, 545, 1961 2026
JWST PRIMER: A deep JWST study of all ALMA-detected galaxies in PRIMER COSMOS – dust-obscured star-formation history back to $z \simeq 7$
3. **Liu, F.-Y.**; Dai, Y. S.; Omont, A.; et al., ApJ, 964, 136 2024
Dust and Cold Gas Properties of Starburst HyLIRG Quasars at $z \sim 2.5$
2. **Liu, F.-Y.**; Dai, Y.; Wu, J.-W.; et al., PrA, 41, 529 2023
CO Emission of Dust-Rich Broad-Emission-Line Quasars
1. **Liu, F.-Y.**; Mai, Y.-F.; Wu, W.-Y.; et al., PhLB, 795, 475 2019
Probing a regular non-minimal Einstein-Yang-Mills black hole with gravitational lensings

Co-authored papers: 11

11. McLeod, D. J.; McLure, R. J.; Dunlop, J. S.; et al., submitted, arXiv: 2604.16666 2026
A search for the first galaxies across $\simeq 0.6$ sq. degrees of JWST imaging: Evidence for a rapid decline in star-formation at $z > 13$
10. Fraser Gillan, A.; Wyrzykowski, L.; Mikołajczyk, P. J.; et al., submitted, arXiv:2603.01383 2026
Time-Domain Photometry and Activity Evolution of Interstellar Comet 3I/ATLAS with BHTOM
9. Taylor, E.; Carnall, A. C.; Maltby, D.; et al., under review of MNRAS, arXiv:2601.02269 2026
The JWST EXCELS survey: Outflows in $1.5 < z < 5$ quiescent galaxies are likely relics from episodic AGN activity
8. Eales, S.; Smith, M.; Bakx, T.; et al., accepted by MNRAS, arXiv:2605.19661 2026
COSMOS-Web: Dust and the hidden star formation in galaxies of different morphological types over the last 13 billion years
7. Leung, H.-H.; Carnall, A. C.; Taylor, E.; et al., accepted by MNRAS, arXiv:2602.0593 2026
The JWST EXCELS survey: The ages and abundances of $3 < z < 5$ massive quiescent galaxies show that downsizing was already in place by $z \simeq 4$
6. Donnan, C. T.; McLeod, D. J.; McLure, R. J.; et al., accepted by ApJ, arXiv:2601.11515 2026
Spectroscopic confirmation of a large and luminous galaxy with weak emission lines at $z = 13.53$
5. Stevenson, S. D.; Carnall, A. C.; Leung, H.-H.; et al., MNRAS, 545, 2087 2025
PRIMER & JADES reveal an abundance of massive quiescent galaxies at $2 < z < 5$
4. Barrufet, L.; Dunlop, J.; Begley, R.; et al., accepted by MNRAS, arXiv:2508.05740 2025
Strength in Numbers: Red Galaxies Bolster the Cosmic Star Formation Rate Density at $z > 3$
3. Donnan, C. T.; McLure, R. J.; Dunlop, J. S.; et al., MNRAS, 533, 3222 2024
JWST PRIMER: a new multifield determination of the evolving galaxy UV luminosity function at redshifts $z \simeq 9 - 15$
2. Stanton, T.; Cullen, F.; McLure, R. J.; et al., MNRAS, 532, 3102 2024
The NIRVANDELS survey: the stellar and gas-phase mass-metallicity relations of star-forming galaxies at $z = 3.5$
1. Hai, X.; Dai, Y. S.; Huang, J.-S.; et al., ApJ, 905, 103 2020
A Large Massive Quiescent Galaxy Sample at $z \sim 1.2$

References

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